Unbounded Gaps Between Semiprimes

An elementary consequence for OEIS A001358

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Abstract

OEIS A001358 lists the integers having exactly two prime factors, counted with multiplicity. We prove that the sequence has unbounded gaps: for every positive integer m there exist infinitely many blocks of m consecutive positive integers containing no semiprime. A Chinese remainder theorem construction forces each integer in such a block to have at least three distinct prime factors. The result is classical in character and no novelty is claimed.

Keywords: semiprimes, prime factors, Chinese remainder theorem, OEIS A001358.

1 Definitions and theorem

Write $\Omega(n)$ for the total number of prime factors of n, counted with multiplicity. OEIS A001358 is precisely the increasing sequence

$$\mathscr{S} = \{n \geq 2 : \Omega(n) = 2\} = \{4, 6, 9, 10, 14, 15, 21, 22, 25, \ldots\}.$$

In particular, prime squares belong to $\mathscr{S}$.

Theorem 1. *For every integer $m \geq 1$, there exist infinitely many positive integers N such that*

$$\boxed{\{N, N+1, \ldots, N+m-1\} \cap \mathscr{S} = \varnothing.} \tag{1}$$

Consequently, if $s_1 < s_2 < \cdots$ are the terms of OEIS A001358, then

$$\limsup_{k\to\infty} (s_{k+1} - s_k) = +\infty.$$

2 Proof

Fix $m \geq 1$. Choose $3m$ distinct primes, grouped as

$$\{p_{j,1}, p_{j,2}, p_{j,3}\}, \qquad j = 0, 1, \ldots, m-1.$$

Set

$$Q_j = p_{j,1} p_{j,2} p_{j,3}, \qquad M = \prod_{j=0}^{m-1} Q_j.$$

The moduli $Q_0, \ldots, Q_{m-1}$ are pairwise coprime. Thus the Chinese remainder theorem yields a residue class $N_0 \pmod{M}$ satisfying all the congruences

$$N_0 \equiv -j \pmod{Q_j} \qquad (0 \leq j < m). \tag{2}$$

Every integer $N = N_0 + tM$ satisfies (2). For each j, all three distinct primes $p_{j,1}, p_{j,2}, p_{j,3}$ divide $N + j$, so for all sufficiently large positive t,

$$\Omega(N + j) \geq 3 > 2 \qquad (0 \leq j < m).$$

Hence none of the m integers $N, N+1, \ldots, N+m-1$ lies in $\mathscr{S}$. As $t \to \infty$, the values of N are unbounded, proving infinitely many blocks.

Finally, $\mathscr{S}$ is infinite: if p varies through odd primes, then $2p$ is a semiprime. For each sufficiently large constructed block, choose consecutive semiprimes $a < b$ enclosing it. Necessarily

$$a < N \leq N + m - 1 < b, \qquad b - a \geq m + 1.$$

Since m was arbitrary, the gaps between consecutive semiprimes are unbounded. □

3 Scope and attribution

This proves a property of the full semiprime sequence, including repeated prime factors such as p^2. It is an elementary Chinese remainder theorem argument and is not presented as a new theorem or a resolution of a named open conjecture. The proof was checked for the stated quantifiers, but it has not received independent peer review.

AI authorship and verification disclosure

Primary author (AI): ChatGPT (OpenAI). The mathematical exposition and preliminary verification were prepared using AI. The note makes no claim of independent certification or originality.

References

[1] OEIS Foundation, *A001358: Semiprimes (or biprimes)*, t https://oeis.org/A001358 (consulted 11 November 2026).

[2] OEIS Foundation, *A001222: Total prime-factor count*, t https://oeis.org/A001222.

[3] The Chinese remainder theorem (classical number-theoretic result).